ICML 2026 · POSTER

Progressive Cramming: Reliable Token Compression and What It Reveals

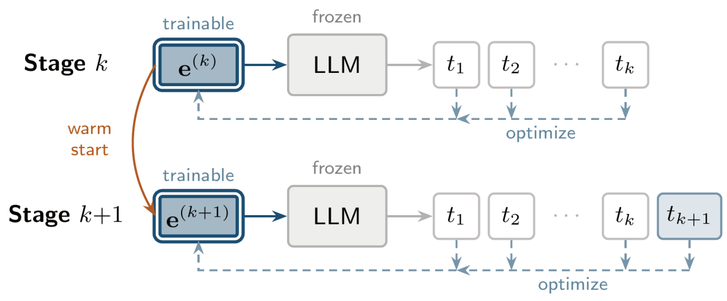
A frozen LM can compress >1000 tokens into one embedding — but perfect reconstruction is brittle steering, not transferable semantics.

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1 Progressive Cramming

Optimize **per-sample** embedding *e* so a **frozen LM** reconstructs tokens. Extend target **one token at a time**, advance only after **perfect** convergence:



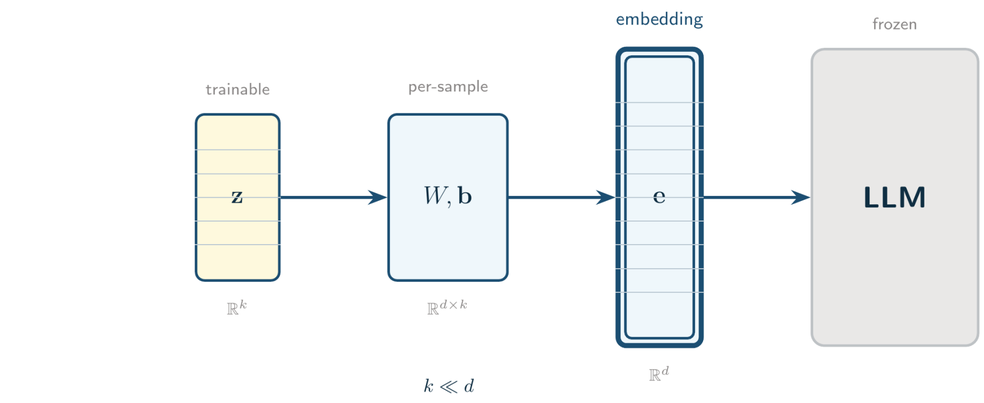
→ Halt on first failure → **sharp capacity boundary**
→ **100% greedy generation** + honest per-sample capacity
→ Saved *e* at each step → **optimizer trajectory**

2 Across Model Families

Mean number of tokens perfectly reconstructed from one embedding on **50 PG19 samples**:

Model	Tokens	+LowDim	Boost
Llama-3.1-8B	1438	**1697**	1.2×
Pythia-1.4b	430	500	1.2×
SmolLM2-1.7B	335	957	2.9×
Gemma-3-4B	214	697	**3.3×**

Low-dim projection. Optimize $z \in \mathbb{R}^k$ with per-sample W, b; $e = Wz + b \in \mathbb{R}^d$, $k \ll d$.

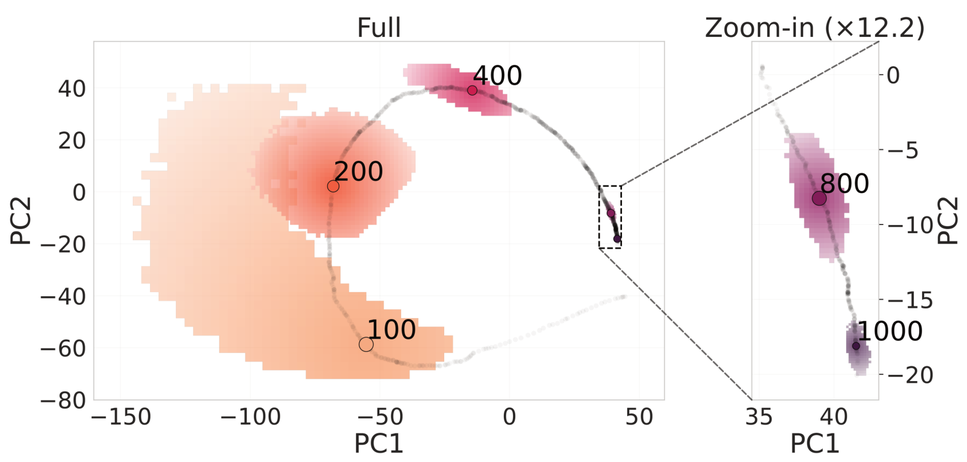


3 Optimisation Trajectories

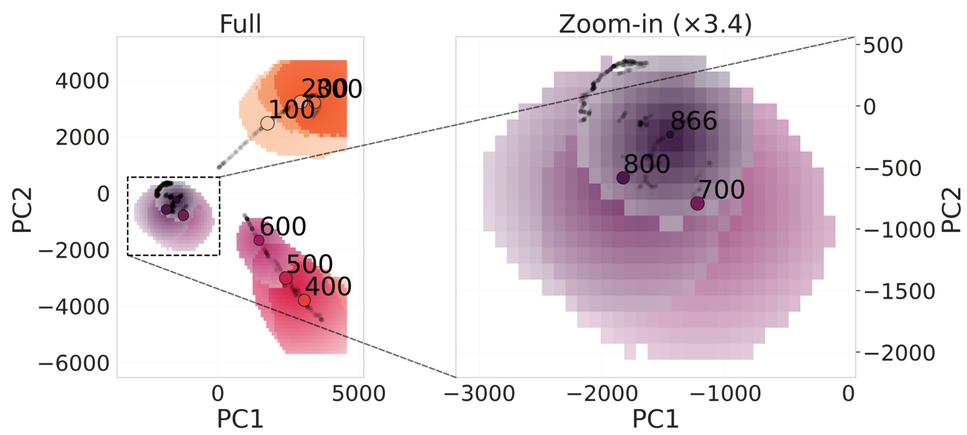
30–100
PCA components for 99% variance

~log
growth with sequence length

→ Embedding trajectories are **intrinsically low-dimensional** despite living in $\mathbb{R}^{4096}$
→ Accuracy basins **shrink** with sequence length — longer sequences leave less room for error



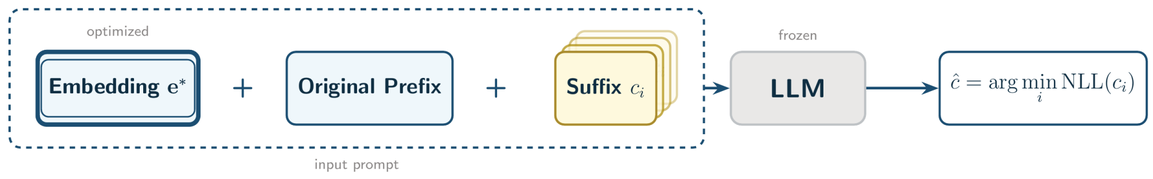
Llama-3.1-8B, PC1–PC2 (65.7% variance). Dots: stage embeddings. Colored regions: reconstruction accuracy at prefixes of 100–1000 tokens.



With low-dim projection (61.0% variance). Colored regions: >90% accuracy at 100-stage intervals. Longer prefixes leave smaller reconstruction basins.

4 Reconstruction ≠ Understanding

Likelihood evaluation



Prepending a crammed embedding **drops** benchmark accuracy, even with the original prefix in context:

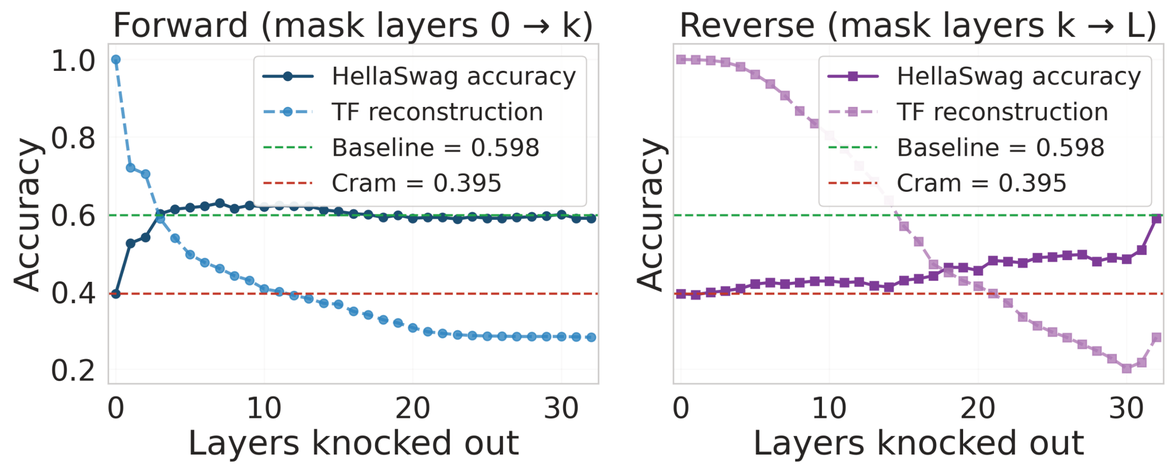
Model	HS Base	HS +Emb	ARC Base	ARC +Emb
Pythia-1.4b	44.0%	37.6%	53.1%	49.1%
SmolLM2-1.7B	53.6%	37.0%	67.3%	56.0%
Llama-3.1-8B	61.9%	40.8%	63.5%	43.3%

HS = HellaSwag; ARC = ARC-E. +Emb prepends the optimized embedding to the original context.

Generative evaluation

On **generative 5-shot MMLU**: accuracy drops to **~0%** with the optimized embedding. A random control stays at baseline.

5 Attention Knockout

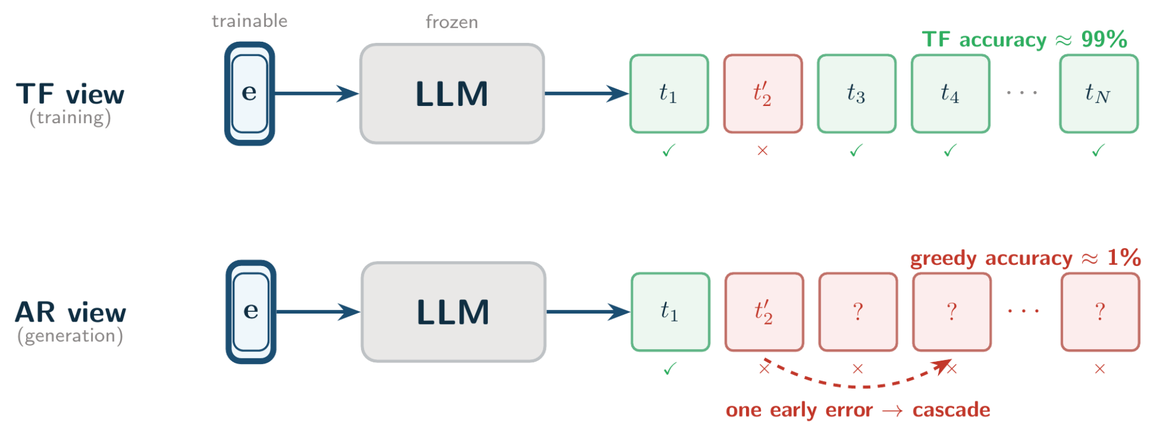


Llama-3.1-8B. Forward (mask early layers) restores downstream accuracy. **Reverse** fails until early layers are reached.

Early layers cause both reconstruction *and* capability collapse. Late-layer attention mass (20–60%) is a symptom. Consistent across Llama-3.1-8B, Pythia-1.4B, SmolLM2-1.7B.

6 Why Not Full Cramming?

Full cramming (Kuratov et al., 2025) fixes the **number of tokens** and optimizes all at once; progressive cramming fixes **reconstruction accuracy** at 100% and grows the sequence. Full cramming reaches higher capacity — but residual errors concentrate on **early positions**, and one early miss cascades through autoregressive generation:



Model / Type	Tokens	TF Acc	Greedy
Llama-8B / Full	1568	99.96%	40.4%
Llama-8B / Progr.	**1438**	**100%**	**100%**
Pythia / Full	512	99.71%	44.2%
Pythia / Progr.	**430**	**100%**	**100%**

7 Limitations & Future Work

→ **Reconstruction ≠ semantics** — we do not yet know how to make a single embedding both perfectly reconstructable and semantically faithful
→ **Optimizer headroom** — low-dim projection results suggest better optimizers could compress longer sequences than we reach

Takeaways

→ **Brittle Steering** — perfect reconstruction does not preserve useful semantics in the tested tasks; downstream benchmarks drop, generative MMLU collapses to ~0%
→ **Early Layers** — across the three tested families, the embedding's causal influence concentrates in the first few layers; late-layer attention mass is a symptom